TQM Newsletter

Issue #15, July 2025

From Editor's desk

Welcome to the 15th edition of the TQM Newsletter!
As we move further into the year, it's encouraging to reflect on the progress we've made in our Total Quality Management journey. Each milestone reflects our shared commitment to excellence, innovation, and continuous improvement. Let's carry this momentum forward with renewed focus and purpose.
We're also excited to introduce a special series on Taiichi Ohno. In this issue – Just-in-time (JIT), Jidoka and Kanban system is detailed out.
Be sure to participate in the TQM Quiz at the end of this issue for a chance to win exciting prizes! Thank you for your continued support. We hope you enjoy this edition as much as we enjoyed creating it.
Happy Reading!

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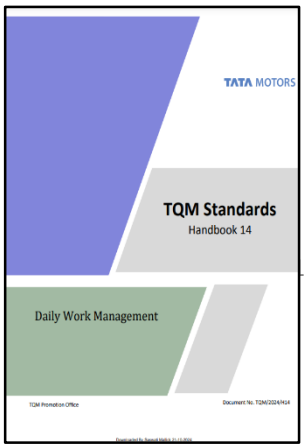
To know more about TQM and various tools used in TQM, kindly refer to the Standards released on TQM portal.

Click here to visit the TQM Portal



Just-in-time (JIT), Jidoka and Kanban system

Continuing with the previous newsletter's focus on Taiichi Ohno's contribution to Lean Manufacturing, this newsletter explores the concepts of Just-in-time (JIT) production, Jidoka and Kanban system.

Just-in-time (JIT) production

Just-in-time is system of production that makes and delivers just what is needed, just when it is needed, and just in the amount needed. JIT aims for the total elimination of all waste to achieve the best possible quality, lowest possible cost and use of resources, and the shortest possible production and delivery lead times.

JIT relies on 'Heijunka' as a foundation, that means "levelization" in Japanese. Heijunka is leveling the type and quantity of production over a fixed period of time. This enables production to efficiently meet customer demands while avoiding batching and results in minimum inventories, capital costs, manpower, and production lead time through the whole value stream.

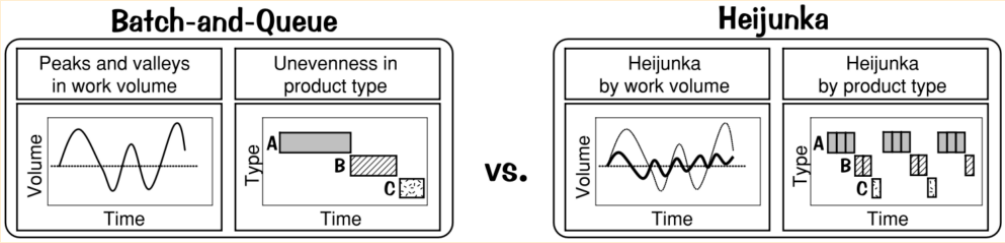


Exhibit.1- Represent Batch- and-Queue vs. Heijunka

Photo credit: lean.org

JIT is comprised of three operating elements: the pull system, takt time, and continuous flow.

Pull system is a method of production control in which downstream activities signal their needs to upstream activities. A downstream operation, whether within the same facility or in a separate facility, provides information to the upstream operation, about what part or material is needed, the quantity needed, and when and where it is needed. Nothing is produced by the upstream supplier process until the downstream customer process signals a need. Pull production strives to eliminate overproduction.

Takt time is a calculation of the available production time divided by customer demand. The purpose of takt time is to precisely match production with demand. It provides the heartbeat of a lean production system.

Continuous flow refers to producing and moving one item at a time (or a small and consistent batch of items) through a series of processing steps as continuously as possible, with each step making just what is requested by the next step. Continuous flow can be achieved in a number of ways, ranging from moving assembly lines to manual cells. It also is called one-piece flow, single-piece flow, and make one, move one.

JIDOKA

Jidoka refers to providing machines and operators the ability to detect when an abnormal condition has occurred and immediately stop work. This enables operations to build in quality at each process and to separate men and machines for more efficient work. Jidoka is one of the two pillars of the Toyota Production System along with just-in-time.

Jidoka sometimes is called 'Autonomation', meaning automation with human intelligence. This is because it gives equipment the ability to distinguish good parts from bad autonomously, without being monitored by an operator. This eliminates the need for operators to continuously watch machines and leads in turn to large productivity gains because one operator can handle several machines, often termed multiprocess handling. Refer exhibit 2.

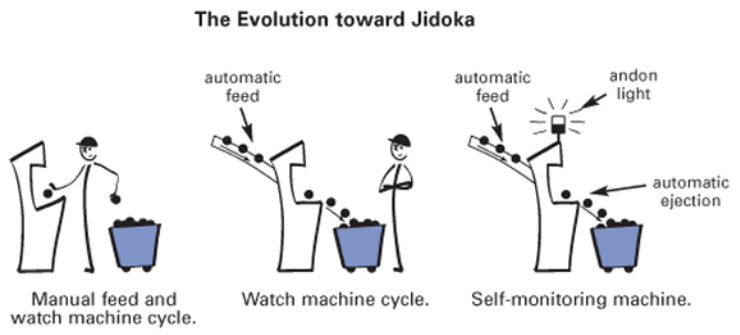


Exhibit.2- Represent Evolution towards Jidoka

Photo credit: lean.org

KANBAN System

A Kanban is a signaling device that gives authorization and instructions for the production or withdrawal (conveyance) of items in a pull system. "Withdrawal" means the conveyance called on by the downstream operation. The term is Japanese for "sign" or "signboard."

Kanban cards are the best-known and most common example of these signals. They often are slips of card stock, sometimes protected in clear vinyl envelopes, stating information such as: part name, part number, external supplier or internal supplying process, pack-out quantity, storage address, consuming process address. A bar code may be printed on the card for tracking or automatic invoicing.

Kanban cards have two functions in a production operation: They instruct processes to make products and they instruct material handlers to move products. The former use is called production kanban; the latter use is termed withdrawal kanban. Exhibit 3 explain the six rule of Kanban.

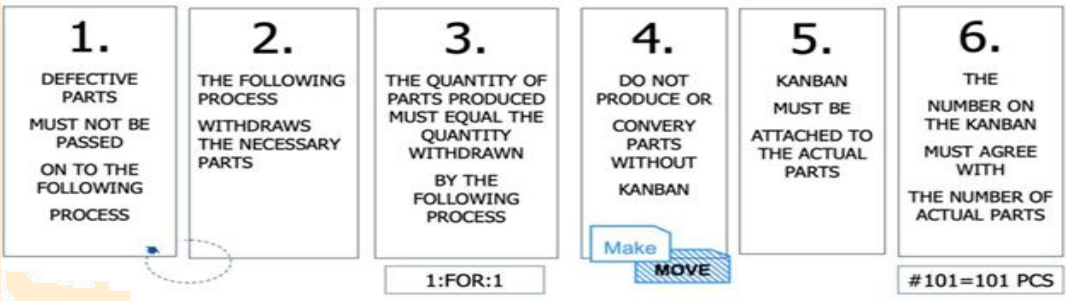
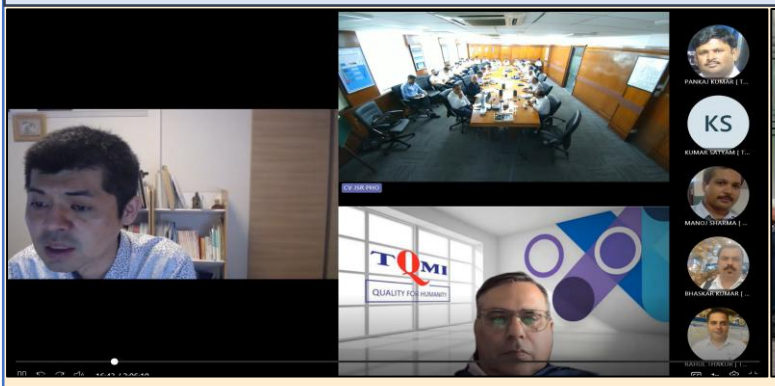


Exhibit 3- Explain Six Rules of Kanban

Glimpse of Key Event in June Month

Mock Diagnosis at JSR by Dr. Sano and Anil Sachdev :



Mock Online assessment in JSR plant on 11th -12th June

Mock assessment was carried out using the finalized setup, which included an additional camera, live streaming, and coverage of the Operational Site Visit (OSV) requirements.

TQMPO Training Session :

TQMPO Training Session on Digital DWM at Gear shop and Paint Factory was conducted by Keshav Mishra.

ASQCT Training by Anil Sachdev :

ASQCT Training Session by Anil Sachdev. In this completion of module 2 for batch -1 was taken. June 19-21

For more details visit TQM Portal, under TQM hub , open "Diagnosis report" and for Diagnosis Question click on "Diagnosis Question Bank"

Best of TQM @ TML

Shining Story @ Lucknow Plant

Objective :Diversity improvement in Rear Axle shop from 18% to 30%
Team Members: Aditi Gupta, Vinod Kumar, Roshni Agnihotri

Steps:

Step 1:Select Theme	Step 2:Define Solution	Step 3:Implmentation Plan	Step 4:Implement	Step 5:Check	Step 6:Standardization	Step 7: Conclusion

Example of TQM tools use : 4W 1H Analysis, Brainstorming Session, Gap Matrix Analysis ,Cause and Effect, GEMBA Observation.

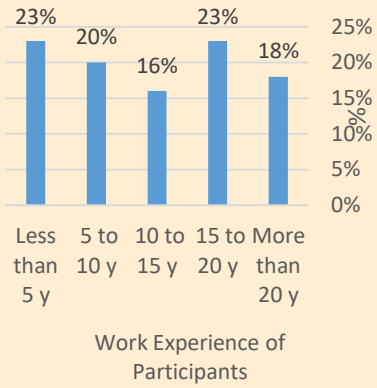
4W 1H Analysis:

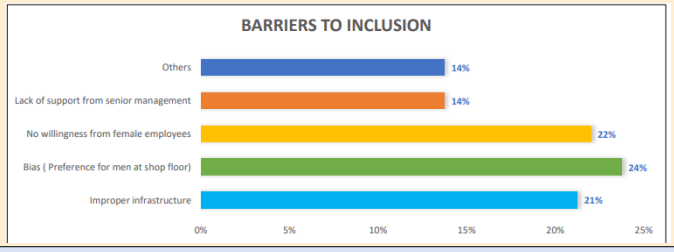
- **What** is the Problem :
  - Low gender diversity (8%)
- **Where** It is :
  - Deployment of female at shop floor
- **When**:
  - Since when :Diversity status at Project start– 8% (Apr '23)
- **Who** is affected :
  - Lucknow plant gender diversity low
- **How** much is the impact?
  - The impact is intangible.
  - With diverse talent, diverse perspectives come along and it reflects organization's commitment to build an inclusive and equitable workspace. Problem solving abilities and innovation at workplace

Data on Diversity-Primary Research-Survey on DEI

Inferences from the survey

- 72% never received a formal training on Diversity , Equity and Inclusion
- Potential barriers to inclusion
- Bias(Preference of men on shop floor)
- Improper infrastructure
- No willingness from female employee





Brainstorming Session:

Reasons of Bias/preference of men at shop floor

- Unavailability of infrastructures supporting female deployment on shop floor.
- Lack of awareness on Diversity , Equity and Inclusion.
- Product challenges of Trucks manufacturing.
- Unavailability of Security provisions.
- Unavailability of Restroom facilities.
- B shift working constraint for females.
- Lack of safety related provisions w.r.t. females.
- Skill gap.
- Lack of inclusive measures for existing female workforce.
- Unavailability of ergonomics considerations in product challenges of trucks.

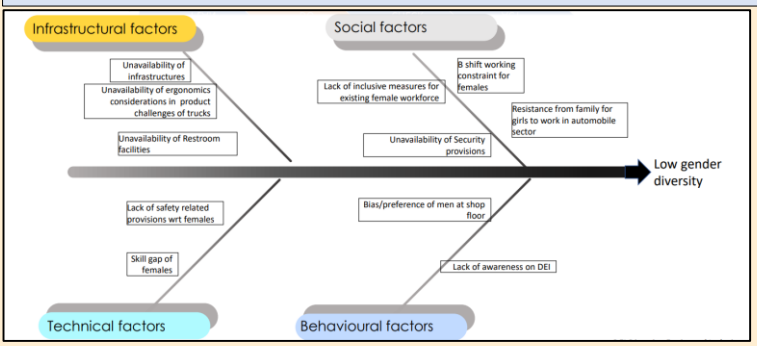
Compiled all the probable causes and their root cause . We found that we lack ground level data on infrastructural gap

Analysis of infrastructure gaps

	Front axle	Rear axle	EV line	IBF	Prolife	BIW	TCF	Paint
Female White collar	✗	✓	✗	✓	✗		✓	✗
Female restroom	✗	○	✓	○	○	✓	○	○
Lady conservancy	✗	○	○		○	○	○	○
Change room with lockers	✗	✗	✗	✗	○	✗	✗	✗
Platforms for Reach issues		○			○	✗	○	✗
Facility modification	○	○	○				○	
Transport for B shift				✗				
CCTV		○	○		○			
First aid training(female)		✗	✗	✗				
	1	7	5	3	5	3	6	2

Quantified the infrastructural analysis and selected rear axle shop as a pilot project for diversity at shop floor.

Identification and Validation of Causes:



SN	Factor	Probable Causes	Cause Validation	Remarks
1	Infrastructure	Unavailability of infrastructures	Gaps in infrastructure observed at certain places. Shop wise analysis not available	Valid
2		Unavailability of ergonomics considerations and product challenges of trucks	Infrastructure- Tools and tackles, weight and height are not designed ergonomically for females	Valid
3		Unavailability of Restroom facilities	Restroom facility available but insufficient in certain shops	Valid
4	Social	Lack of inclusive measures for existing female workforce	Representation of females in plant sub-committees , self directed team, union members is low	Valid
5		Unavailability of Security provisions	Security team is available but for security check at gate , female security guards are required	Invalid
6		B shift working constraint for females	B shift working of females started with door to door dropping by cab at shift end	Invalid
7	Behavioural	Bias/preference of men at shop floor	Supervisors and managers prefer men over women for deployment at shop floor	Valid
8		Lack of awareness on DEI	Survey conducted – 72% responded that they have received no formal training on Diversity Equity and Inclusion	Valid
9	Technical	Lack of safety related provisions wrt females	No specific training on female working on shop floor	Invalid
10		Skill gap for females	No skill gap since the hiring criteria is common for both genders	Invalid

Categorized the probable causes into 4 broad factors namely social, infrastructure, behavioral, technical. Factors which came out were validated.

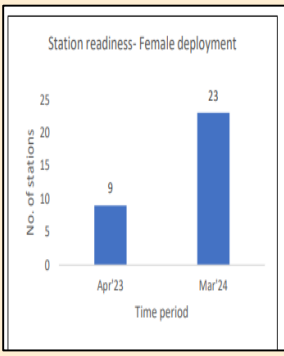
Actions:

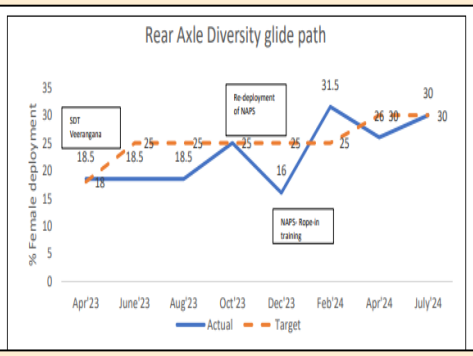
OPERATION	CONSTRAINT FOR FEMALE DEPLOYMENT	COUNTER MEASURE	RESULT
Beam washing operation	Height constraint for female deployment. Without height support, an unsafe situation created for female deployment	Platforms provided for safe deployment of female workforce	Deployment of female workforce

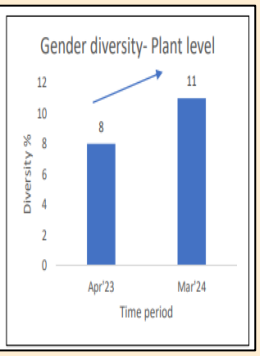
BEFORE

AFTER

Effect:







a) Female deployment Readiness of rear axle stations increased from 20.5 % to 52% from Apr'23 to Mar'24

b) Absolute figures: No. of females in Apr'23 : 13
No. of females in Mar'24 : 32

c) Absolute figures: No. of females in Apr'23 : 239
No. of females in Mar'24 : 390

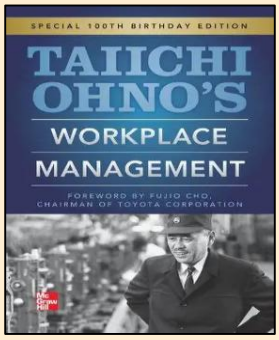
Learn and Participate

Book of the Month

TAIICHI OHNO'S:WORKPLACE MANAGEMENT

Published: 1988

Ohno delivers timeless lessons on how to effectively manage the gemba actual place or work. He relates stories from across his nearly 40 years of struggle to establish the Toyota Production System as both a mindset and supporting behaviors of constant improvement...



Curiosity Corner

Sakichi Toyoda: The founder of Toyoda looms, he invented a loom that would automatically stop if a thread broke. This concept of building in a "human touch" to automation became a key pillar of TPS, known as **Jidoka**.

Winner of Last Quiz

Pankaj Thaman (TMCV, Press Factory) from Pune Plant and Brijesh Singh (TMCV, HR, Skill Dev.) from LKO, declared as the winner of TQM Newsletter Quiz 14TH edition.

Participate in the quiz to win exciting prizes !!

Link for the quiz- https://forms.office.com/r/AzxAy93MiF